
International skin spectra archive dataset and statistical analysis

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Abstract

1 International skin spectra archive dataset

The International Skin Spectra Archive (ISSA) [Lu et al., 2025] is a multicultural dataset of human skin phenotypes comprising 15,256 spectral and colorimetric records from 2,113 participants. The measurements were collected between 2012 and 2024 and originate from 11 component datasets contributed by laboratories in eight countries. All measurements were acquired under a standardized protocol, which facilitates consistency across datasets.

Skin spectral reflectance refers to the wavelength-dependent proportion of incident light reflected by the skin. It is governed by interactions between light and the layered tissue structure and chromophore composition of the skin [Anderson and Parrish, 1981, Jacques, 2013]. For the present analysis, we extracted reflectance measurements over the visible wavelength range from 400 to 700 nm at 10-nm intervals, yielding 31 spectral measurements per record. The measurements were obtained from 12 anatomical sites: the back of the hand, cheek, cheekbone, chin, earlobe, forehead, inner arm, neck, nose tip, outer arm, palm, and ring finger.

For each skin reflectance curve, we estimated the first derivative using a Savitzky–Golay filter [Schafer, 2011] with a window length of five wavelength points and a third-order polynomial. Figure 1 presents the skin reflectance curves and their corresponding first derivatives after outlier removal.

The spectral properties of human skin have long been studied in research on visual perception and human evolution. Human skin is a multilayered organ whose visible color is influenced primarily by absorption due to melanin in the epidermis and hemoglobin in the blood, together with wavelength-dependent light scattering within the tissue [Igarashi et al., 2007]. To summarize melanin-related attenuation and hemoglobin-related absorption, we calculated the melanin index (MI) and erythema index (EI) for each reflectance spectrum:

$$\text{MI} = 100 \log_{10} \left(\frac{1}{R_{680}} \right), \quad (1)$$

$$\text{EI} = 100 \log_{10} \left(\frac{R_{680}}{R_{560}} \right), \quad (2)$$

where R_{680} and R_{560} denote the reflectance values at 680 and 560 nm, respectively.

To characterize variation in human skin color, we used the five-category eumelanin human skin color scale proposed by Dadzie et al. [2022]. Each reflectance spectrum was assigned to one of the following categories according to its melanin index: eumelanin low (EML), $\text{MI} < 25$; eumelanin intermediate-low (EMIL), $25 \leq \text{MI} < 50$; eumelanin intermediate (EMI), $50 \leq \text{MI} < 75$; eumelanin intermediate-high (EMIH), $75 \leq \text{MI} < 100$; and eumelanin high (EH), $\text{MI} \geq 100$.

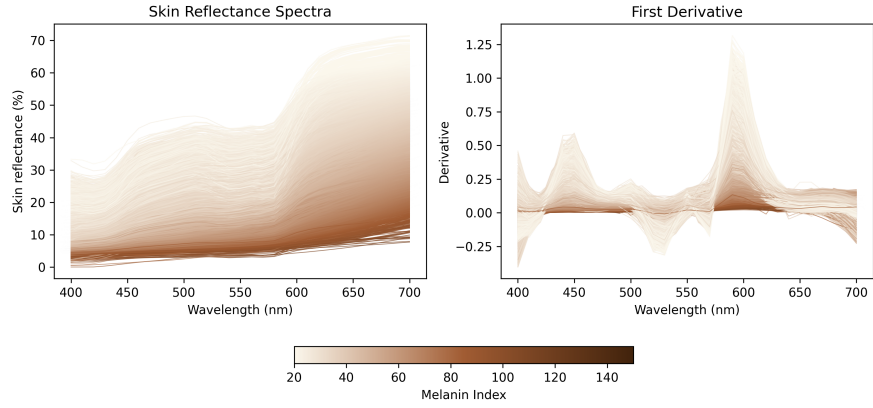


Figure 1: ISSA skin reflectance curves and their corresponding first derivatives after outlier removal.

Figure 2 shows the ISSA skin reflectance curves and their first derivatives stratified by eumelanin class. The magnitude of the first derivative generally decreased as the melanin index increased, indicating that the reflectance curves tended to become flatter with increasing skin pigmentation.

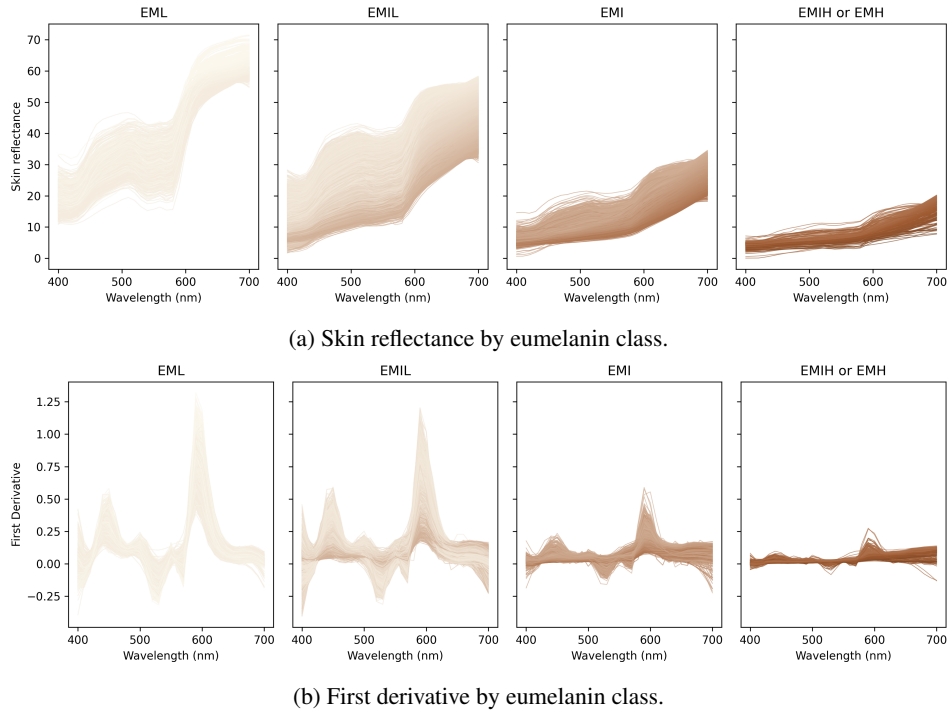


Figure 2: ISSA skin reflectance curves and their first derivatives stratified by eumelanin class.

2 Derivative-based shape analysis

Haemoglobin, the red chromophore, introduces the distinctive W shape in the skin reflectance spectra [Angelopoulou, 1999]. To characterize the local shape variation, we conduct a derivative-based analysis. The analysis focus on the wavelength range from 500 to 600 nm, where the spectra displayed substantial variation in curvature and local absorption features. In this analysis, we focus on the six facial sites and the palm, where facial sites includes cheek, cheek bone, chin, ear lobe, forehead and nose tip.

Let N_i denote the number of negative-to-positive derivative crossings detected between 500 and 600 nm. Such a sign change corresponds to a local minimum in the original reflectance curve. The spectra were classified into three principal shape categories:

$$\text{Shape}_i = \begin{cases} \text{Increasing,} & N_i = 0, \\ \text{U-shape,} & N_i = 1, \\ \text{W-shape,} & N_i = 2. \end{cases}$$

Spectra with more than two detected minima were treated as unclear or noisy. The increasing category represented spectra with no detected valley and a predominantly positive slope over the selected wavelength range. A U-shaped spectrum contained one local valley, whereas a W-shaped spectrum contained two distinct valleys.

Figure 3 shows the distributions of the three spectral shapes across anatomical sites. W-shaped spectra were most common on the palm, accounting for 537 of 769 spectra (69.8%). In contrast, U-shaped spectra were dominant at most facial sites. They accounted for 75.9% of the nose-tip spectra, 64.3% of the chin spectra, 49.6% of the forehead spectra, and 48.2% of the cheekbone spectra. The cheek exhibited a more balanced distribution: 26.1% of its spectra were W-shaped, 38.3% were U-shaped, and 35.6% were increasing. These results indicate that local spectral shape depends strongly on the anatomical measurement site.

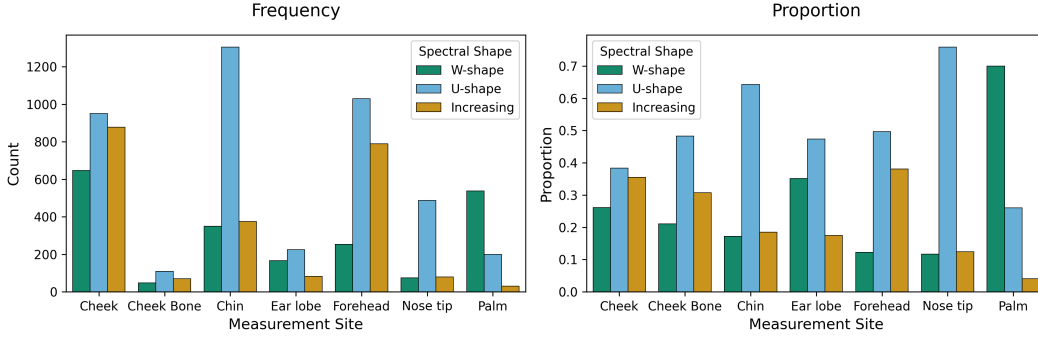


Figure 3: ISSA spectral shape distributions at each body site.

Spectral shape also varied systematically with the melanin index, as shown in Figure 4. Among spectra in the eumelanin-low class, W-shaped and U-shaped profiles were predominant, whereas increasing profiles were uncommon. For example, only 2.6% of eumelanin-low forehead spectra and 5.1% of eumelanin-low cheek spectra were classified as increasing. No eumelanin-low palm spectrum exhibited an increasing profile. The proportion of W-shaped spectra generally decreased as the melanin index increased. U-shaped spectra were most prevalent in the eumelanin-intermediate-low class at several facial sites, including the chin (72.1%) and nose tip (78.5%). At higher melanin-index values, increasing spectra became more common. Within the eumelanin-intermediate class, the increasing shape accounted for 96.9% of cheek spectra, 94.2% of forehead spectra, 83.3% of cheekbone spectra, and 62.6% of chin spectra. The palm followed a somewhat different pattern: W-shaped spectra remained dominant in both the eumelanin-low and eumelanin-intermediate-low classes, accounting for 78.0% and 69.7% of the spectra, respectively.

Overall, lighter skin spectra tended to exhibit W-shaped or U-shaped profiles, while spectra with higher melanin indices increasingly displayed a smooth, increasing profile. However, the relationship between melanin index and spectral shape varied across anatomical sites. In particular, the palm showed a much higher prevalence of W-shaped spectra than the facial sites. These findings suggest that derivative-based spectral shape captures variation associated with both skin pigmentation and anatomical location.

3 Bayesian varying-coefficient model

To characterize the wavelength-dependent associations between skin reflectance and major skin chromophores, we develop a Bayesian varying-coefficient model (BVCM). The analysis is restricted

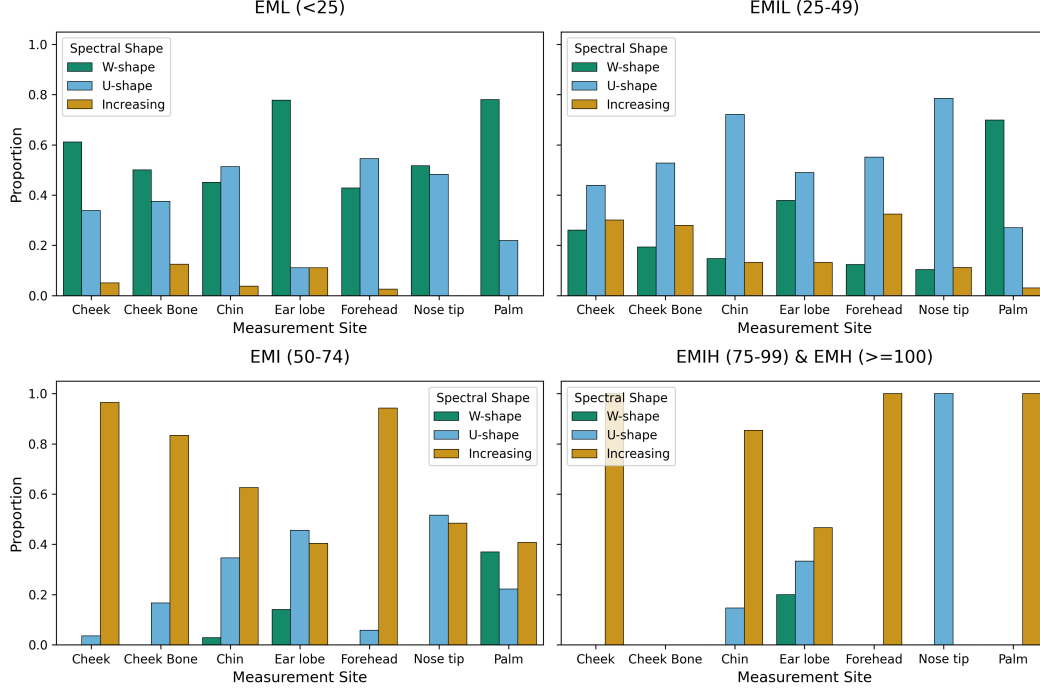


Figure 4: ISSA spectral shape distributions across eumelanin classes at each body site.

to spectra measured at six facial sites: the cheek, cheekbone, chin, earlobe, forehead, and nose tip. After preprocessing and derivative-based outlier removal, the data were randomly divided into training and test sets containing 80% and 20% of the spectra, respectively.

Let $R_i(\lambda) \in (0, 1)$ denote the reflectance of spectrum i at wavelength λ , where $\lambda \in [400, 700]$ nm. We apply the logit transformation

$$Y_i(\lambda) = \text{logit}\{R_i(\lambda)\} = \log \left\{ \frac{R_i(\lambda)}{1 - R_i(\lambda)} \right\},$$

and model the transformed reflectance as

$$Y_i(\lambda) = \beta_0(\lambda) + X_{\text{MI},i}\beta_{\text{MI}}(\lambda) + X_{\text{EI},i}\beta_{\text{EI}}(\lambda) + Z_i(\lambda) + \epsilon_i(\lambda), \quad (3)$$

where $X_{\text{MI},i}$ and $X_{\text{EI},i}$ denote the melanin and erythema indices, respectively; $\beta_0(\lambda)$ is a wavelength-dependent intercept; and $\beta_{\text{MI}}(\lambda)$ and $\beta_{\text{EI}}(\lambda)$ are wavelength-dependent coefficient functions associated with the melanin and erythema indices. The function $Z_i(\lambda)$ represents a subject-specific deviation from the population-level reflectance curve. The measurement errors are assumed to be independent and identically distributed as

$$\epsilon_i(\lambda) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\epsilon^2).$$

Figure 5 shows the joint distribution of the melanin and erythema indices across the six facial sites.

3.1 Varying coefficient functions

The intercept and chromophore coefficient functions are represented using cubic B-spline bases. For $k \in \{0, \text{MI}, \text{EI}\}$, we write

$$\beta_k(\lambda) = \sum_{j=1}^J b_{k,j} B_j(\lambda), \quad J = 20, \quad (4)$$

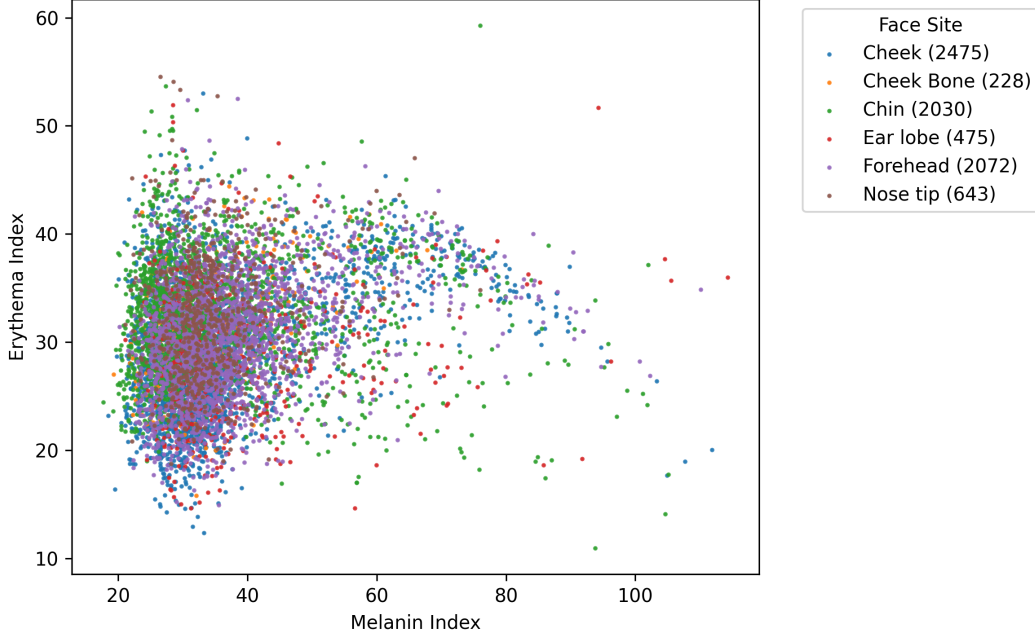


Figure 5: Relationship between the melanin and erythema indices across facial measurement sites.

where $B_j(\lambda)$ is the j th B-spline basis function and $b_{k,j}$ is the corresponding coefficient. Smoothness is induced through a second-order random-walk prior,

$$\Delta^2 b_{k,j} = b_{k,j} - 2b_{k,j-1} + b_{k,j-2} \sim \mathcal{N}(0, \tau_\beta^2), \quad (5)$$

with $\tau_\beta^2 = 1$. This prior penalizes abrupt changes between neighboring spline coefficients while allowing the effects of melanin and erythema to vary smoothly with wavelength.

3.2 Subject-specific deviations

The subject-specific deviation functions are assigned a zero-mean Gaussian process prior,

$$Z_i(\lambda) \sim \text{GP} \{0, K_Z(\lambda, \lambda')\}, \quad (6)$$

where K_Z is a Matérn-5/2 covariance kernel with length-scale parameter ρ_Z and amplitude parameter η_Z [Williams and Rasmussen, 2006]. These random functions capture residual spectral variation that is not explained by the melanin and erythema indices. For computational efficiency, the Gaussian process is approximated using a finite Hilbert-space representation.

The Gaussian-process hyperparameters and residual variance are assigned the following priors:

$$\rho_Z \sim \text{Gamma}(50, 1), \quad (7)$$

$$\eta_Z \sim \text{HalfNormal}(0.5), \quad (8)$$

$$\sigma_\epsilon^2 \sim \text{InverseGamma}(1, 1). \quad (9)$$

3.3 Posterior inference

Posterior inference is performed using a hybrid Gibbs sampler. Parameters with conditionally conjugate posterior distributions are updated using Gibbs steps, whereas the Gaussian-process hyperparameters are updated using a Metropolis step. We ran 10,000 tuning iterations followed by 10,000 retained posterior iterations. The acceptance rate for the hyperparameter update was 15.83%.

Figure 6 presents the posterior estimates of the varying coefficient functions, including their posterior means and pointwise 95% credible intervals.

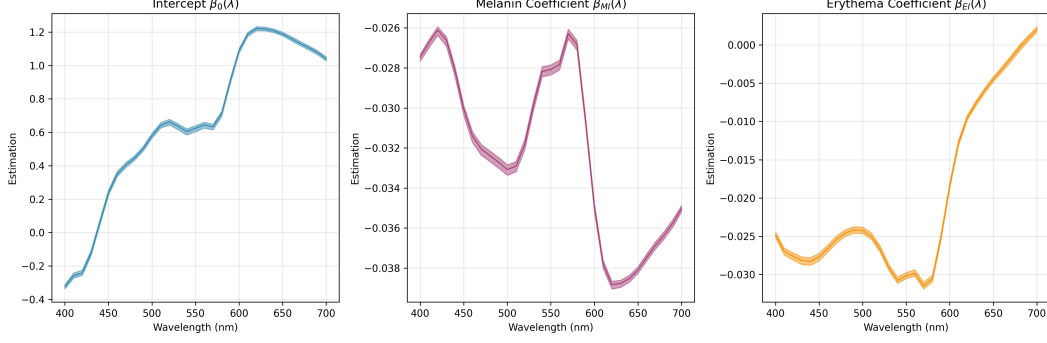


Figure 6: Posterior estimates of the varying coefficient functions. Solid lines represent posterior means, and shaded regions represent pointwise 95% credible intervals.

For a given pair of melanin and erythema indices, the posterior population-level reflectance curve is calculated as

$$\hat{R}(\lambda \mid X_{\text{MI}}, X_{\text{EI}}) = \text{logit}^{-1} \left\{ \hat{\beta}_0(\lambda) + X_{\text{MI}} \hat{\beta}_{\text{MI}}(\lambda) + X_{\text{EI}} \hat{\beta}_{\text{EI}}(\lambda) \right\}. \quad (10)$$

As shown in Figure 7, the predicted curves exhibit a consistent change in spectral shape as the melanin index increases across low, median, and high values of the erythema index. Spectra with low melanin indices tend to exhibit a W-shaped profile between approximately 500 and 600 nm, whereas spectra with higher melanin indices show a smoother and predominantly increasing profile. This pattern is consistent with the derivative-based shape analysis.

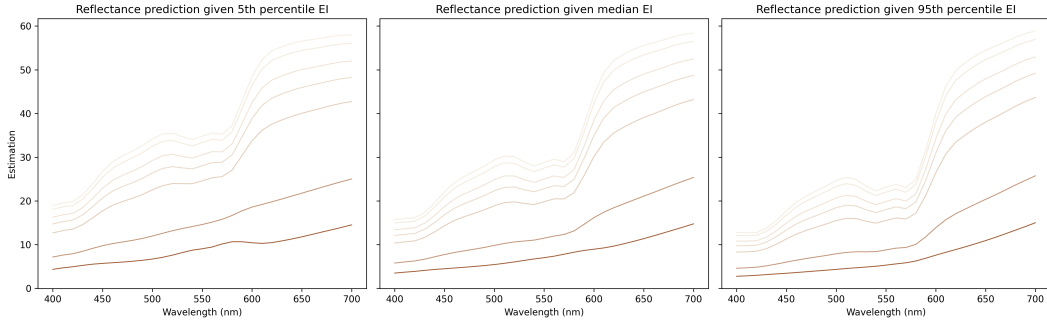


Figure 7: Population-level reflectance curves predicted by the BVCM for selected melanin and erythema index values.

3.4 Model evaluation

We evaluate population-level predictions using the varying coefficient functions alone and assess in-sample reconstruction after incorporating the estimated subject-specific deviations. Table 1 summarizes the resulting predictive performance.

The population-level component explains approximately 96.5% of the variation in both the training and test sets, with a test RMSE of 1.427 percentage points. The similar performance on the training and test sets indicates that the estimated wavelength-dependent chromophore effects generalize well to held-out spectra. Incorporating the subject-specific deviation functions substantially improves in-sample reconstruction by capturing individual variation around the population-level curves. For held-out subjects, predictions are based on the population-level component because their subject-specific deviation functions are not observed during model fitting.

We further compare the predicted reflectance curves with the observed curves in the test set. Figure 8 shows that the BVCM captures the main shapes and wavelength-dependent patterns of the held-out reflectance spectra well.

Table 1: Predictive performance of the Bayesian varying-coefficient model on the ISSA dataset. RMSE and MAE are reported in percentage points of reflectance.

Dataset	Model component	R^2	RMSE	MAE
Training	Population-level fixed effects	0.9653	1.410	1.052
Training	Fixed effects and subject-specific deviations	0.9993	0.198	0.132
Test	Population-level fixed effects	0.9646	1.427	1.068

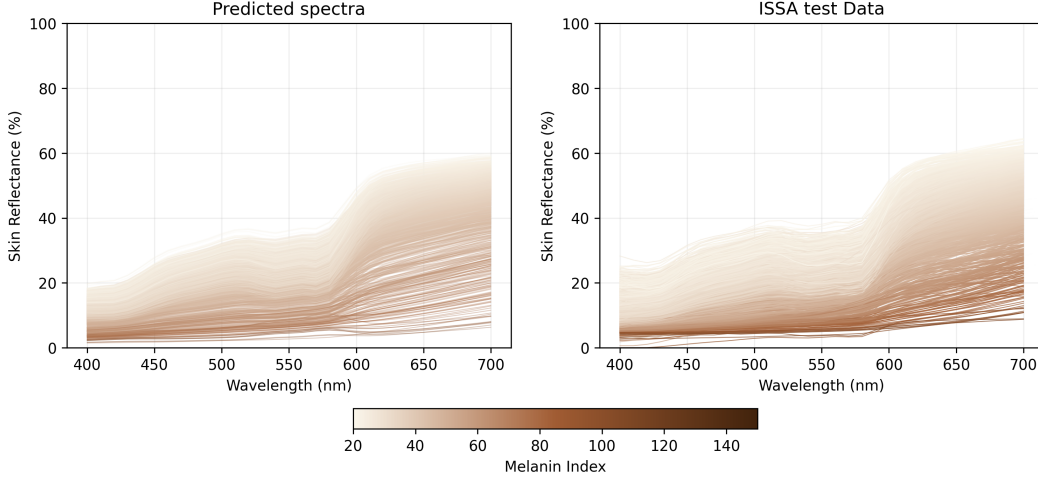


Figure 8: Comparison of the BVCM predictions with the observed reflectance spectra in the ISSA test set.

4 Skin Reflectance Validation Framework

Generating realistic skin reflectance spectra is fundamental to research in computer graphics. Aliaga et al. [2023] proposed BioSkin, a biophysical model that generates skin reflectance spectra from a set of skin-property parameters. They further developed an encoder–decoder framework to characterize the relationships among skin properties, spectral reflectance, and RGB color. However, an important question remains: how can we ensure that the generated reflectance spectra are realistic? A closely related question is how the skin-property parameters should be constrained to biologically plausible ranges.

Our BVCM provides a statistical framework for validating simulated skin reflectance spectra and identifying parameter combinations that are consistent with empirical observations. In particular, we develop a two-stage validation procedure. The first stage evaluates whether the melanin and erythema indices of a candidate spectrum are consistent with their joint distribution in the ISSA data. The second stage assesses the plausibility of the entire reflectance curve conditional on these indices.

4.1 Chromophore Index Validation

We model the joint distribution of the melanin and erythema indices in the ISSA facial-site data using a five-component Gaussian mixture model:

$$p(x) = \sum_{k=1}^5 \pi_k \mathcal{N}(x; \mu_k, \Sigma_k), \quad x = \begin{pmatrix} X_{\text{MI}} \\ X_{\text{EI}} \end{pmatrix}, \quad (11)$$

where π_k , μ_k , and Σ_k denote the mixing probability, mean vector, and covariance matrix of component k , respectively.

A candidate spectrum passes the first validation stage if its melanin and erythema indices fall within the 95% probability region of at least one Gaussian component. This criterion removes spectra whose chromophore-index combinations are inconsistent with those observed in the ISSA population.

Figure 9 shows the fitted Gaussian mixture model and the corresponding 95% probability regions for the melanin and erythema indices in the ISSA facial-site data.

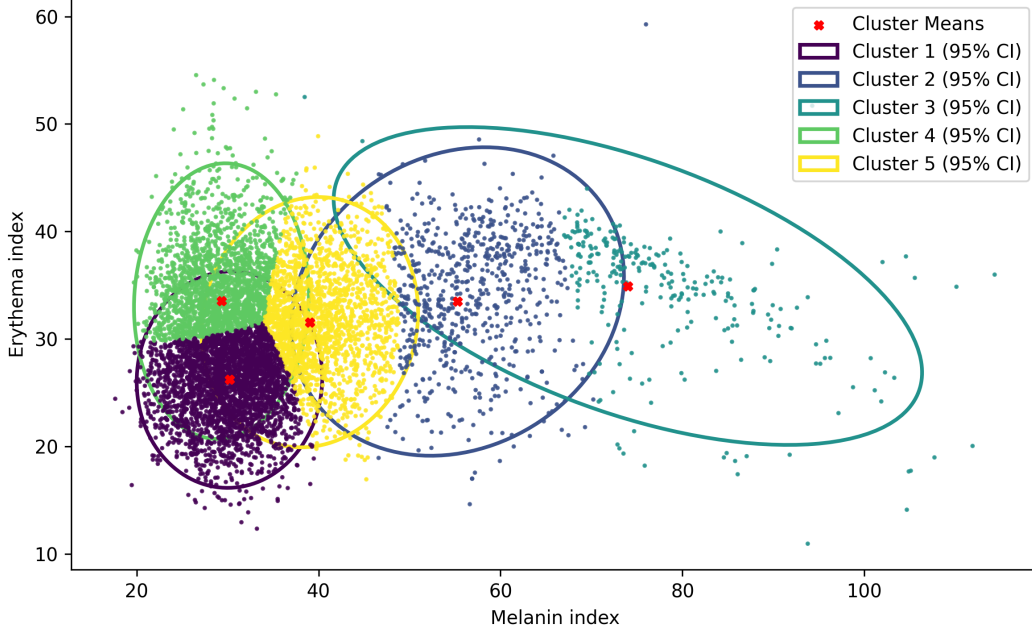


Figure 9: Five-component Gaussian mixture model fitted to the melanin and erythema indices in the ISSA facial-site data. Ellipses indicate the 95% probability regions of the mixture components.

4.2 Spectral Plausibility Assessment

The second validation stage assesses whether the complete spectral shape of a candidate is consistent with the fitted BVCM. For candidate spectrum i , the BVCM population-level prediction on the logit scale is

$$\hat{Y}_i = \hat{\beta}_0 + X_{\text{MI},i}\hat{\beta}_{\text{MI}} + X_{\text{EI},i}\hat{\beta}_{\text{EI}}, \quad (12)$$

where each coefficient vector contains its estimated values across the observed wavelengths. The candidate spectrum is transformed to the logit scale as

$$Y_i = \text{logit}(R_i),$$

and the corresponding residual vector is defined as

$$e_i = Y_i - \hat{Y}_i.$$

The covariance matrix of the residual spectral variation is estimated from the subject-specific Gaussian process and measurement-error components of the BVCM:

$$\Sigma_{\text{total}} = \hat{K}_Z + \hat{\sigma}_\epsilon^2 I, \quad (13)$$

where $\hat{K}_Z$ is the estimated covariance matrix of the subject-specific deviations. We then calculate the spectral Mahalanobis distance

$$D_i = (e_i^\top \Sigma_{\text{total}}^{-1} e_i)^{1/2}. \quad (14)$$

Rather than relying on a theoretical chi-squared approximation in the high-dimensional spectral space, we obtain an empirical threshold from the ISSA training data. Specifically, the validation threshold is defined as the 95th percentile of the Mahalanobis distances calculated for the training spectra:

$$c_{0.95} = Q_{0.95}(D_1^{\text{train}}, \dots, D_{n_{\text{train}}}^{\text{train}}). \quad (15)$$

A candidate spectrum is classified as spectrally plausible if $D_i \leq c_{0.95}$.

4.3 Validation Results

We apply the proposed validation framework to skin reflectance spectra generated by BioSkin. Figure 10 summarizes the results of the two-stage validation procedure.

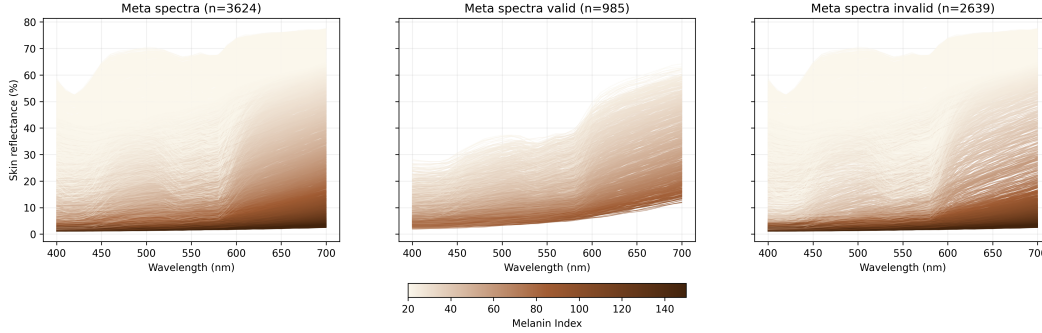


Figure 10: Results of the two-stage BVCM validation procedure applied to skin reflectance spectra generated by BioSkin.

Table 2 summarizes the numerical validation results. Overall, 986 of the 3,624 candidate spectra (27.2%) pass both validation stages, whereas 2,638 spectra (72.8%) are rejected. The retained spectra have melanin indices ranging from 20.37 to 97.38 and erythema indices ranging from 16.32 to 49.44.

Table 2: Results of the two-stage validation procedure applied to the skin reflectance spectra generated by BioSkin.

Validation outcome	Number of spectra	Percentage
Initial candidate spectra	3,624	100.0%
Passed chromophore-index validation	1,481	40.9%
Rejected by chromophore-index validation	2,143	59.1%
Passed spectral plausibility assessment	986	66.6% of Stage 1
Rejected by spectral plausibility assessment	495	33.4% of Stage 1
Passed both validation stages	986	27.2%
Rejected overall	2,638	72.8%

The substantial rejection rate indicates that broad sampling of the parameters of a biophysical skin model does not necessarily produce spectra that resemble empirical human skin reflectance. The first stage removes candidates with implausible combinations of pigmentation- and blood-related indices. The second stage further identifies spectra whose local shapes or wavelength-dependent variation are inconsistent with the ISSA data, even when their summary indices are plausible. The retained spectra therefore satisfy both the population-level chromophore constraints and the full-curve spectral constraints learned from observed human skin measurements.

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